

Experience

[Amazon Web Services](#) /Seattle, WA

APR 20 – PRESENT

Software Development Engineer II [OCT2022-PRESENT]

- **Next gen patching:**
Responsibilities: Proposed and led the next gen MSK (Managed streaming Kafka) patching to eliminate downtime and customer impact during maintenance. Coordinated across teams/leadership and product to gather consensus and meet customer expectation. Mentored and guided junior engineers on the team to build this new solution.
Achievements: Delivered a new distributed patching system that eliminated maintenance window and made clusters always available, while improving the patching time from once a month to twice a month.
- **Custom billing solution:**
Responsibilities: Led the design, planning, and implementation of a custom billing system for AWS's capacity reservations. Designed a credit calculation engine with seconds-level granularity across 27 regions and 80 Availability Zones, and developed a scalable distributed feed generator producing ~150K billing files per month with daily reconciliation to ensure data integrity. Coordinated delivery with four cross-functional teams across three time zones.
Achievements: Successfully launched custom billing solution for AWS's new capacity reservation feature which was essential to renew contracts for some of big AWS customers like stripe, Untold, technicolor etc.
- **AWS capacity reservation notification feature:**
Responsibilities: Led the design and implementation of a notification system for underutilized capacity reservations. This system processed millions of EC2 pool entries by dynamically scaling number of lambda functions depending on the pool utilization. Used CDK to create required infrastructure such as IAM roles, S3 buckets, lambdas, step functions etc. Designed a different retry mechanism to counter dependency/network failure. Built a dashboard for leadership which showed various metrics such as number of underutilized reservations, number of notifications sent, internal vs external customers etc.
Achievements: Successfully implemented AWS capacity reservation notification feature which proactively notified customers through email, personal health dashboard and cloud watch event for their underutilized capacity reservations. AWS what's new post [link](#)

Software Development Engineer [APR2020-OCT2022]

- **Cloud Formation Integration:**
Responsibilities: Led efforts to integrate capacity reservation service with cloud formation. Worked with doc writers, security engineers and legal team for compliance and release. Responsible for design and implementation of this integration. Created all the infrastructure like pipelines, accounts, environments, alarms, internal repository and external git hub repository. Implemented all the handlers required by the cloud formation integration.
Achievements: Created a dashboard which was reviewed weekly with the team to monitor the health of our service. Git hub repository [link](#)
- **Fraud Service integration:**
Responsibilities: Headed integration with fraud team to delete resources from fraudulent accounts. Ensured that this integration didn't impact legitimate customers. Created a testing environment by mocking fraud service to test the flow end to end. Tested various scenarios which gave confidence.
Achievements: In 2022, saved approximately \$90-100K by releasing fraudulent reservations.

[University at Buffalo](#)/NY

MAY 19 - AUG 2019

Research assistant

- **Responsibilities:** Worked with a research team on identifying feasibility rendering of AR objects under TEE (Trusted Execution Environment) funded by National Science Foundation. Designed and implemented an application to map pixel by pixel data to Linux Frame buffer (fb0 device). Built an Android NDK app that displays open GL frames which ran on Hikey and Nitrogen board.
Achievements: Research team published a paper on displaying images on TEE. [Link](#)

Education

[Masters in Computer Science](#) / [University at Buffalo] / GPA – 3.7/4.0

AUG 18 - DEC 19

[Bachelor of Engineering](#) / [K. J. Somaiya College, Mumbai] / GPA – 7.4/10.0

JULY 14 - MAY 17

Additional Achievements

- [Game Development](#)/Unity, C#
Developed an android game on unity. It is an extension of the famous flappy bird game. In this game, user can choose different characters which have different weight and speed, also user can set the difficulty level according to his/her choice. Play store [link](#)
- [AR \(Augmented Reality\) Development](#)/Android, AR-CORE
Developed an AR game for Android using Google ARCORE platform. It is simple game where balls pop out of the surface and user has to click the balls to get score if user misses a ball, it decrements the score by 1. User should avoid hitting the special ball, which leads to game over. Also, if score gets below 0 it leads to game over. Play store [link](#).